

Youngseok Lee

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Education

Sungkyunkwan University, Nano Engineering

Experience

NORMA Inc., Quantum AI Developer, Quantum AI Team

Research and development of quantum machine learning models and the classical simulation tools used to train them.

- Co-developed and maintain PADO-Pauli, a GPU-accelerated differentiable Pauli-propagation surrogate engine (Python package padopauli).
- Proposed and analyzed mixed IQP-QCBMs, ancilla-extended IQP generative models with provable barren-plateau avoidance and a cluster initialization scheme.
- Developed SAFE ma-QAOA, a surrogate-assisted, parameter-distilled multi-angle QAOA pipeline with exact fine-tuning.
- Applied QGAN and IQP-QCBM models to molecular generation for KRAS G12D inhibitor candidates, including experiments on Quantinuum H2 and IonQ Forte.
- Contributed to an in-house quantum SDK (circuit representation, user API, circuit output and visualization).

Publications

Trainability and Mode Separation of Mixed IQP-QCBMs

Youngseok Lee, Hyunwoo Kim

arxiv.org/abs/2607.27883

SAFE ma-QAOA: Surrogate-Assisted and Fine-Tuning Enhanced Multi-Angle QAOA with Parameter Distillation

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PADO-Pauli: A GPU-Accelerated Implementation of Pauli Propagation

Hyunwoo Kim, Youngseok Lee

github.com/Norma-Q/PADO-Pauli

Talks

- **SAFE ma-QAOA**, contributed talk (accepted), IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026.
- **PADO-Pauli**, poster (accepted), IEEE Quantum Week 2026 (QCE26), Toronto, Canada, September 13–18, 2026.
- **Trainability and Mode Separation of Mixed IQP Circuits**, contributed talk, [ML4QT Symposium](#), Institute for Quantum Computing, University of Waterloo, September 23–25, 2026.

Patents

- **Quantum Computing Method Using a Pauli Graph Generated by Zero Filtering to Perform Surrogate Operations and Quantum Computing System Performing the Same.** Korean patent, application no. 10-2026-0070495 (filed 2026-04-20), registered. Applicant: NORMA Inc.
- **Quantum Artificial Intelligence Training Method Performing Surrogate Training Using Pauli Graphs and Quantum Artificial Intelligence Training System Performing the Same.** Korean patent, application no. 10-2026-0070494 (filed 2026-04-20), registered. Applicant: NORMA Inc.

Research Mentoring

- Supervised research intern Dohyun Kim (January 19 – April 19, 2026) on *Reinforcement Learning for Adaptive Pauli Propagation*, and wrote a letter of recommendation.

Skills

Quantum computing

Software

Interests

Research interests